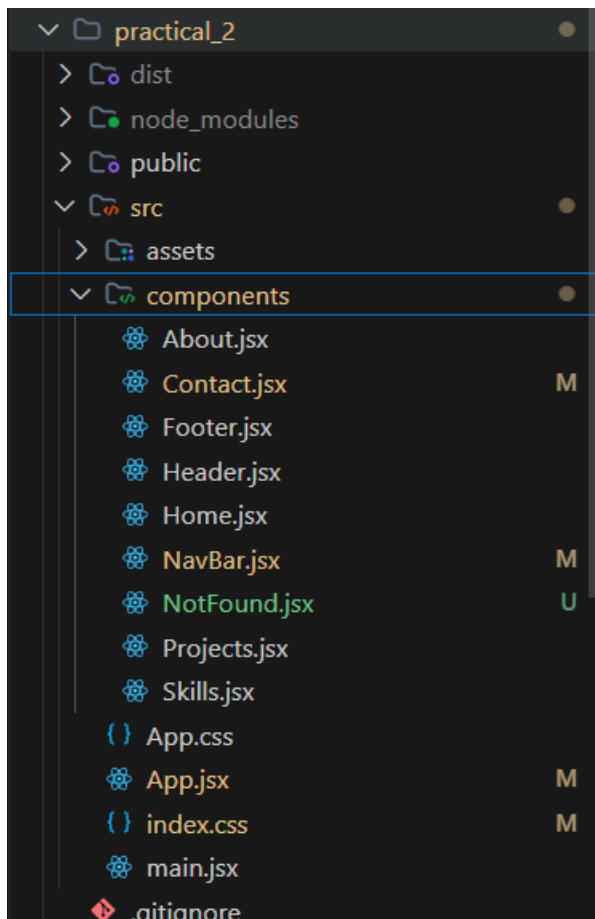


Practical 2: State Management and Routing in React

Problem Statement

- Extend the portfolio application from Practical 1 by adding React Router.
- Implement a navigation bar with at least 3 routes: Home, Projects, and Contact.
- Each route must render a distinct component without a full page reload.
- Add at least 2 useState variables used meaningfully one for toggling UI visibility and one for managing a form input on the Contact page.
- The Contact page must include a controlled input that captures and displays user input in real time.

Project Structure:



Source Code Screenshots:

App.jsx

```
1 import { BrowserRouter as Router, Routes, Route } from "react-router-dom";
2 import NavBar from "../components/NavBar";
3 import Home from "../components/Home";
4 import Projects from "../components/Projects";
5 import Contact from "../components/Contact";
6 import Footer from "../components/Footer";
7
8 function App() {
9   const studentName = "Het Rakholiya";
10
11   return (
12     <Router>
13       <div className="container">
14         <NavBar />
15
16         <main className="main-content">
17           <Routes>
18             <Route path="/" element={<Home />} />
19             <Route path="/projects" element={<Projects />} />
20             <Route path="/contact" element={<Contact />} />
21           </Routes>
22         </main>
23
24         <Footer name={studentName} />
25       </div>
26     </Router>
27   );
28 }
29
30 export default App;
```

NavBar.jsx

```
1 import { NavLink } from "react-router-dom";
2
3 function NavBar() {
4   return (
5     <nav className="navbar">
6       <div className="navbar-logo">
7         Het's Portfolio
8       </div>
9       <ul className="navbar-links">
10        <li>
11          <NavLink
12            to="/"
13            className={({ isActive }) => isActive ? "nav-link active" : "nav-link"}
14            end
15          >
16            Home
17          </NavLink>
18        </li>
19        <li>
20          <NavLink
21            to="/projects"
22            className={({ isActive }) => isActive ? "nav-link active" : "nav-link"}
23          >
24            Projects
25          </NavLink>
26        </li>
27        <li>
28          <NavLink
29            to="/contact"
30            className={({ isActive }) => isActive ? "nav-link active" : "nav-link"}
31          >
32            Contact
33          </NavLink>
34        </li>
35      </ul>
36    </nav>
37  );
38 }
39
40 export default NavBar;
```

Home.jsx

```
1 import { useState } from "react";
2 import About from "../About";
3 import Skills from "../Skills";
4
5 function Home() {
6   const [showMoreInfo, setShowMoreInfo] = useState(false);
7
8   const skilllist = [
9     "React JS",
10    "JavaScript",
11    "HTML",
12    "CSS",
13    "Node JS"
14  ];
15
16  return (
17    <div className="home-container">
18      <About />
19
20      <div className="toggle-section">
21        <button
22          className="btn-toggle"
23          onClick={() => setShowMoreInfo(!showMoreInfo)}
24        >
25          {showMoreInfo ? "Hide Profile Details" : "Show Profile Details"}
26        </button>
27
28        {showMoreInfo && (
29          <div className="more-info-card fade-in">
30            <h3>Additional Details</h3>
31            <p>
32              Beyond development, I enjoy contributing to open-source, learning cloud architectures, and solving algorithmic challenges.
33            </p>
34            <p><strong>Current Goal:</strong> Mastering full-stack development using modern web frameworks.</p>
35          </div>
36        )}
37      </div>
38
39      <Skills skills={skilllist} />
40    </div>
41  );
42 }
43
44 export default Home;
```

Projects.jsx

practical_2 > src > components > Projects.jsx > [⌘] default

```
1 function Projects() {
2   const projects = [
3     {
4       id: 1,
5       title: "Calibration Management System",
6       description: "A web application for tracking equipment calibration schedules, managing certificate logs, and automated maintenance reminders.",
7       tech: ["React", "Node.js", "Express", "PostgreSQL"],
8       link: "#"
9     },
10    {
11      id: 2,
12      title: "Quotation Management System",
13      description: "A comprehensive tool to create, track, and generate professional PDF sales quotations and invoices with real-time tax calculation.",
14      tech: ["React", "JavaScript", "Tailwind CSS", "Node.js"],
15      link: "#"
16    },
17    {
18      id: 3,
19      title: "Android Service Book Offline App",
20      description: "An offline-first mobile app for field service engineers to log maintenance records, job sheets, and service history without internet connectivity.",
21      tech: ["Android", "Kotlin", "SQLite", "Room DB"],
22      link: "#"
23    }
24  ];
25
26  return (
27    <section className="projects-container">
28      <h2>My Projects</h2>
29      <p className="projects-sub">A curated selection of the projects I've built recently.</p>
30      <div className="projects-grid">
31        {projects.map((project) => (
32          <div key={project.id} className="project-card">
33            <h3>{project.title}</h3>
34            <p>{project.description}</p>
35            <div className="project-tags">
36              {project.tech.map((tag, idx) => (
37                <span key={idx} className="tech-tag">{tag}</span>
38              ))}
39            </div>
40            <a href={project.link} className="project-link" onClick={(e) => e.preventDefault()}>
41              View Details &rarr;
42            </a>
43          </div>
44        ))}
45      </div>
46    </section>
47  );
48 }
49
50 export default Projects;
```

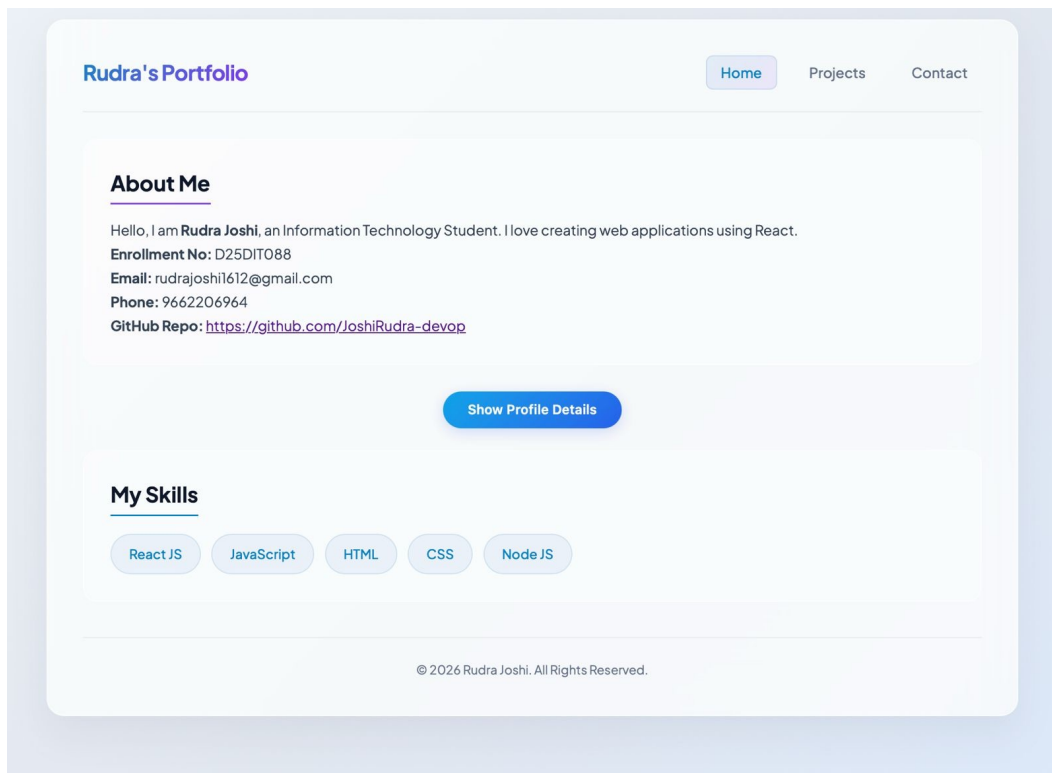
Contact.jsx

```

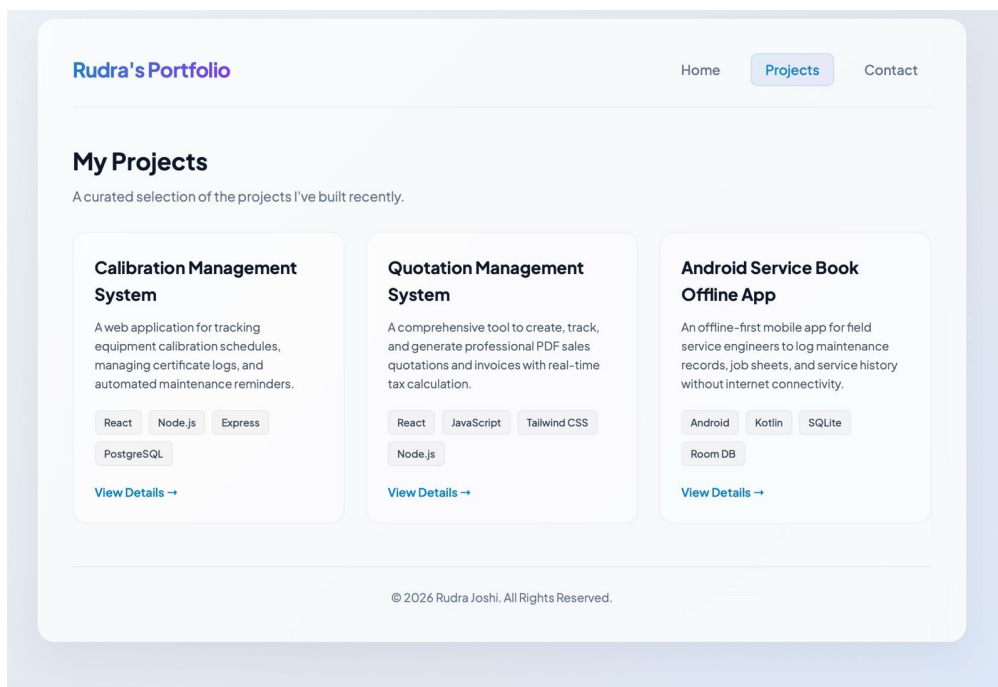
1 import { useState } from "react";
2
3 function Contact() {
4   const [formData, setFormData] = useState({
5     name: "",
6     email: "",
7     message: ""
8   });
9   const [submitted, setSubmitted] = useState(false);
10
11   const handleChange = (e) => {
12     const { name, value } = e.target;
13     setFormData((prevData) => ({
14       ...prevData,
15       [name]: value
16     }));
17   };
18
19   const handleSubmit = (e) => {
20     e.preventDefault();
21     if (formData.name && formData.email && formData.message) {
22       setSubmitted(true);
23     }
24   };
25
26   const handleReset = () => {
27     setFormData({ name: "", email: "", message: "" });
28     setSubmitted(false);
29   };
30
31   return (
32     <section className="contact-container">
33       <h2>Contact Me</h2>
34       <p className="contact-sub">Have a question or want to work together? Let's connect!</p>
35
36       <div className="contact-layout">
37         {submitted ? (
38           <div className="submission-success fade-in">
39             <h3>Thank You, {formData.name}!</h3>
40             <p>Your message has been sent successfully. I will get back to you at <strong>{formData.email}</strong> as soon as possible.</p>
41             <button className="btn-reset" onClick={handleReset}>Send Another Message</button>
42           </div>
43         ) : (
44           <form className="contact-form" onSubmit={handleSubmit}>
45             <div className="form-group">
46               <label htmlFor="name">Full Name</label>
47               <input
48                 type="text"
49                 id="name"
50                 name="name"
51                 value={formData.name}
52                 onChange={handleChange}
53                 placeholder="Enter your name"
54                 required
55               />
46             </div>
57             <div className="form-group">
47               <label htmlFor="email">Email Address</label>
58               <input
59                 type="email"
60                 id="email"
61                 name="email"
62                 value={formData.email}
63                 onChange={handleChange}
64                 placeholder="Enter your email"
65                 required
66               />
67             </div>
68             <div className="form-group">
69               <label htmlFor="message">Message</label>
70               <textarea
71                 id="message"
72                 name="message"
73                 value={formData.message}
74                 onChange={handleChange}
75                 placeholder="Write your message here..."
76                 rows="5"
77                 required
78               />
79             </div>
80             <button type="submit" className="btn-submit">Send Message</button>
81           </form>
82         )}
83
84       <div>
85         <div>
86           <div>
87             <div>
88               <div>
89                 <div>
90                   <div>
91                     <div>
92                       <div>
93                         <div>
94                           <div>
95                             <div>
96                               <div>
97                                 <div>
98                                   <div>
99                                     <div>
100                                       <div>
101                                         <div>
102                                           <div>
103                                             <div>
104                                               <div>
105                                                 <div>
106                                                   <div>
107                                                     <div>
108                                                       <div>
109                 </div>
110               </div>
111             </div>
112           </div>
113         </div>
114       </div>
115     </section>
116   );
117 }
118
119 export default Contact;

```

Output Screenshots:



Home Route showing user profile and interactive details toggle



Projects Route displaying curated static portfolio cards

Rudra's Portfolio

Home Projects **Contact**

Contact Me

Have a question or want to work together? Let's connect!

Full Name

Email Address

Message

Send Message

Real-time Input Preview

NAME:
Not entered yet

EMAIL:
Not entered yet

MESSAGE:
Start typing to see preview...

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Contact Route showing responsive contact form and real-time preview card

Supplementary Problems

1. Custom 404 Error: Configured a NotFound.jsx component and registered a wildcard route path='*' in App.jsx to render this page for unknown paths.
2. Dark/Light Mode: Created an isDarkMode state in App.jsx, toggled via a navbar icon. Used useEffect to dynamically apply the dark-mode class to the root element, with dark style rules added to index.css.
3. Character Count: Integrated state tracking in the contact form and added a counter displaying the length of the message field in real-time.

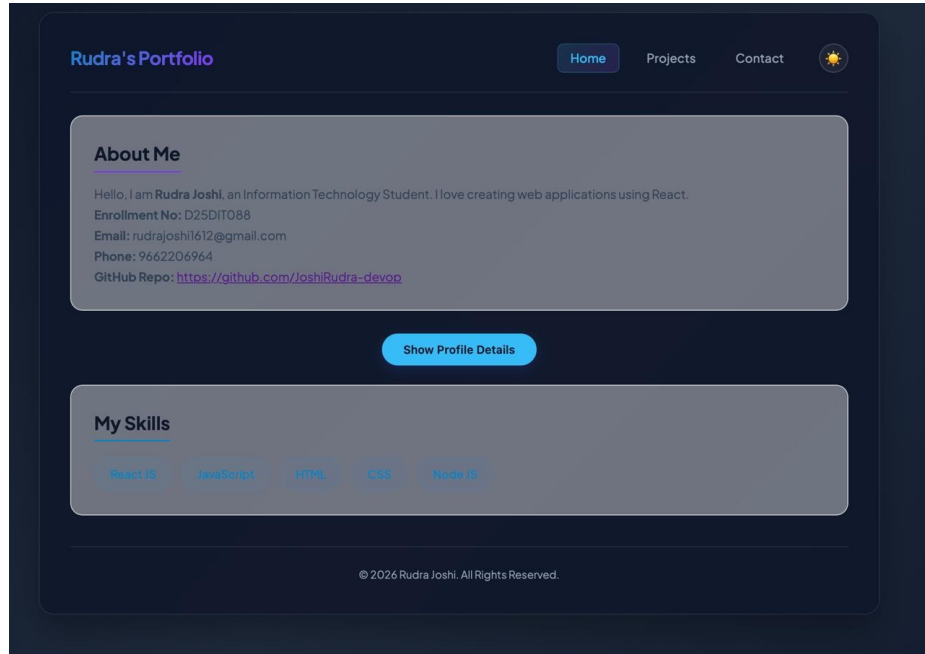
Source Code of NotFound.jsx

```
import React from "react";
import { Link } from "react-router-dom";

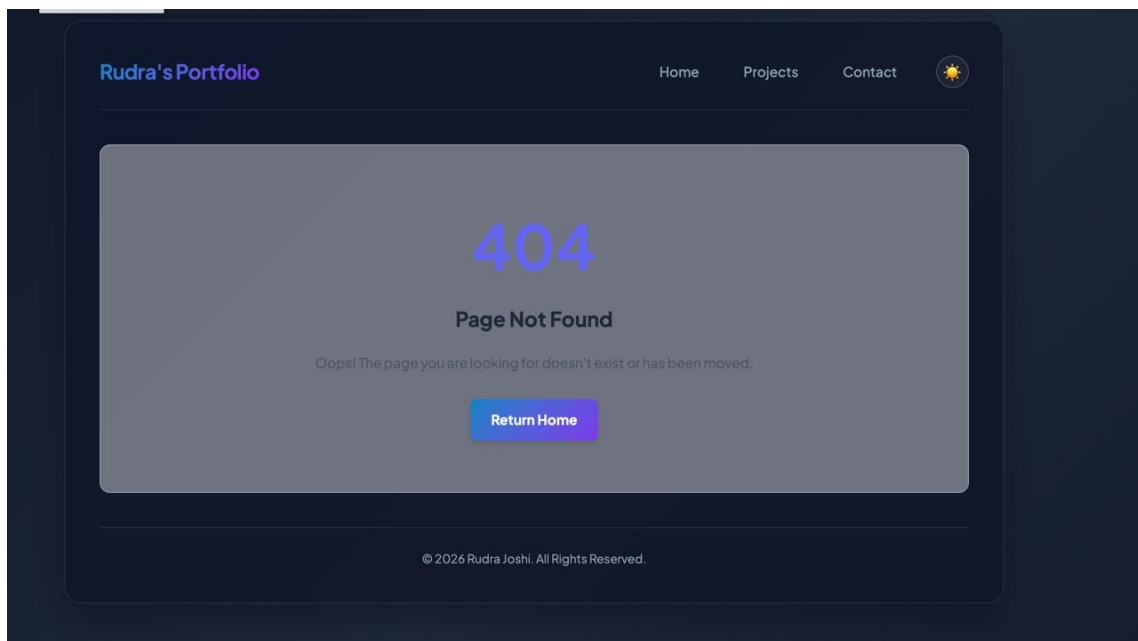
function NotFound() {
  return (
    <div style={{
      textAlign: "center", padding: "60px 20px", background: "rgba(255, 255, 255, 0.4)",
      borderRadius: "12px", border: "1px solid rgba(255, 255, 255, 0.3)", backdropFilter:
"blur(10px)", margin: "40px 0"
    }}>
      <h2 style={{ fontSize: "72px", margin: "0", color: "#6366f1" }}>404</h2>
      <h3 style={{ fontSize: "24px", margin: "10px 0", color: "#1f2937" }}>Page Not Found</h3>
      <p style={{ margin: "20px 0 30px", color: "#4b5563" }}>
        Oops! The page you are looking for doesn't exist or has been moved.
      </p>
      <Link to="/" style={{
        padding: "12px 24px", background: "linear-gradient(135deg, #0284c7 0%, #7c3aed
100%)",
        color: "ffffff", textDecoration: "none", fontWeight: "bold", borderRadius: "8px",
        boxShadow: "0 4px 6px rgba(0, 0, 0, 0.1)", display: "inline-block"
      }}>
        Return Home
      </Link>
    </div>
  );
}

export default NotFound;
```

Output Screenshots of Supplementary Problems



Toggled dark mode styles active on the portfolio's main layout.



Customized 404 error template rendered on entering an invalid route.

Rudra's Portfolio[Home](#)[Projects](#)[Contact](#)

Contact Me

Have a question or want to work together? Let's connect!

Full Name

Email Address

Message

Hi this is rudra joshi and I am testing the contact page to real time text preview

Character Count: 82

Send Message

Real-time Input Preview

NAME:
Rudra JOshi

EMAIL:
rudrajoshi1612@gmail.com

MESSAGE:
Hi this is rudra joshi and I am testing the contact page to r
eal time text preview

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Live controlled character count displayed under the message input field.